

Skills Summary

Picking a Method, and Reading the Roots

Use this reference while completing your worksheet or when studying.

Skill 1 — When the squared part is already alone

Use **square roots** when the equation looks like $(\text{something})^2 = k$. Get the square alone, root both sides, write $\pm$ straight away, then undo whatever is inside.

Do *not* expand $(x - 2)^2$ first — that turns two steps into five.

Example: Solve $(x - 2)^2 = 9$.

Step 1. The squared bracket is already by itself and there is no other x in the equation, so square roots are the shortest method.

Step 2. $x - 2 = \pm 3$, so $x = 2 + 3$ or $x = 2 - 3$.

$$x = -1 \text{ or } x = 5$$

Try it: Solve $(x + 1)^2 = 4$.

$$x = -3 \text{ or } x = 1$$

Skill 2 — Factor, then use the zero product

Use **factoring** when one side is 0 and the other factors quickly. Then $A \cdot B = 0$ forces $A = 0$ or $B = 0$.

If there is no constant term, the common factor is x — factor it out, never divide by it, or the root $x = 0$ is lost.

Example: Solve $x^2 - x - 12 = 0$.

Step 1. One side is already zero and the leading coefficient is 1, so a quick product-and-sum search is faster than the formula.

Step 2. Product -12 , sum -1 : the pair is -4 and 3 , so $(x - 4)(x + 3) = 0$.

$$x = -3 \text{ or } x = 4$$

Try it: Solve $x^2 + 6x + 8 = 0$.

$$x = -2 \text{ or } x = -4$$

Skill 3 — When nothing factors: the formula

Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

It always works, so use it when a factor pair does not appear within a few seconds. Write a , b , c with their signs first; the denominator is $2a$, not 2.

Example: Solve $x^2 - 2x - 4 = 0$; give the larger root to two decimal places.

Step 1. No pair of whole numbers multiplies to -4 and adds to -2 , so factoring is a dead end and the formula is the method.

Step 2. $a = 1$, $b = -2$, $c = -4$, so $-b = 2$ and the discriminant is $4 + 16 = 20$: $x = \frac{2 + \sqrt{20}}{2}$ for the larger root.

$$x \approx 3.24$$

Try it: Solve $x^2 + x - 3 = 0$; give the larger root to two decimal places.

check: $x \approx 1.30$

Skill 4 — Reading the roots back into the story

Two questions, in order: first solve the equation and keep *every* root; then ask which roots make sense for the quantity in the story. A root that fails the story is still a correct solution of the equation — it is rejected for meaning, not for arithmetic.

Example: A stone is thrown up from the ground; after t seconds its height in feet is $-16t^2 + 64t$. When does it land?

Step 1. Landing means the height is zero, so solve the height equation for t and only then decide which root answers the question.

Step 2. $-16t(t - 4) = 0$, so $t = 0$ or $t = 4$. Both are real times; $t = 0$ is the throw itself.
 $t = 0$ or $t = 4$, and the stone lands after 4 seconds.

Try it: Same throw, but the height is $-16t^2 + 48t$. Solve for t .

check: $t = 0$ or $t = 3$